



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-[Section] System Integration and Architecture

System Design Document (SDD)

Project Title: Campus Bites

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EXECUTIVE SUMMARY

1.1 Project Overview

Campus Bites is a simplified campus-based food ordering marketplace designed to allow students and faculty to browse food products, add items to a cart, and place orders within the campus environment.

The system consists of:

- Spring Boot backend API
- React Web Application
- Android Mobile Application
- PostgreSQL database

The goal is to provide a fast, simple, and secure ordering platform tailored specifically for campus users.

1.2 Objectives

1. Develop a fully functional food ordering MVP for campus users
2. Implement a secure authentication system using JWT
3. Provide product browsing, cart management, and order placement
4. Enable administrators to manage food items and monitor orders
5. Ensure seamless integration across backend, web, and mobile platforms

1.3 Scope

Included Features

- User registration and authentication
- Food product listing with search functionality
- Shopping cart management
- Checkout and order placement
- Admin panel for food item management
- Order tracking (basic status view)
- Responsive web interface

- Native Android mobile app
- MongoDB relational database

Excluded Features

- Online payment gateway integration
- Real-time delivery tracking
- Push notifications
- Email notifications
- Loyalty rewards system
- Advanced analytics dashboard

1.0 INTRODUCTION

1.1 Purpose

This System Design Document (SDD) defines the overall system architecture, functional requirements, non-functional requirements, and implementation plan for Campus Bites.

It serves as a reference document for instructors, developers, and stakeholders to ensure alignment during development and deployment.

2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

2.1 Project Overview

Project Name: Campus Bites

Domain: Campus Food Ordering / E-commerce

Primary Users:

1. Customers (Students, Faculty, Staff)
2. Administrators (Canteen Managers / System Admins)

Problem Statement

Students and campus staff need a fast and convenient way to browse food options and place orders without physically lining up at food stalls. Manual ordering causes long queues, delays, and inefficient order management.

Proposed Solution

Campus Bites is a web and mobile-based ordering platform that allows users to:

- Register and log in securely
- Browse available food items
- Add items to cart
- Place orders with delivery or pickup options
- Track order status
- Allow administrators to manage menu items and orders

The system follows a three-tier architecture consisting of:

- Frontend (React Web + Android Mobile)
- Backend (Spring Boot REST API)
- Database (PostgreSQL/MySQL)

2.2 Core User Journeys

Journey 1: First-Time Customer Order

1. User opens the web or mobile application
2. Clicks **Sign Up**
3. Enters required details (name, email, password)
4. System validates and creates account
5. User is automatically logged in
6. User browses available food items
7. User selects item and views details
8. User adds item to cart
9. User proceeds to checkout
10. User confirms order
11. System displays order confirmation

Journey 2: Returning Customer Order

1. User logs in using email and password
2. User browses or searches for food items
3. User adds multiple items to cart
4. User updates item quantities
5. User proceeds to checkout
6. User confirms order
7. User receives order confirmation

Journey 3: Administrator Management

1. Admin logs in using admin credentials
2. Admin navigates to dashboard
3. Admin adds new food items
4. Admin updates price or stock
5. Admin deletes unavailable items
6. Admin views incoming orders
7. Admin updates order status (Preparing, Ready, Completed)

2.3 Feature List (MoSCoW Prioritization)

MUST HAVE

1. User registration
2. User login and logout
3. JWT-based authentication
4. Food item listing
5. Food item details view
6. Add to cart functionality
7. Update/remove cart items
8. Checkout and order placement
9. Admin product CRUD
10. Order management dashboard

SHOULD HAVE

1. Search functionality
2. Order history
3. Order status tracking
4. Input validation feedback
5. Responsive design

COULD HAVE

1. Favorite food items
2. Sales summary dashboard
3. Filter by category

WON'T HAVE

1. Online payment gateway integration
2. Email notifications
3. Push notifications
4. Social media login
5. Advanced analytics

2.4 Detailed Feature Specifications

Feature: User Authentication

- **Screens:** Registration, Login, Forgot Password
- **Fields:** Email, Password, Confirm Password, First Name, Last Name
- **Validation:** Email format, password strength (min. 8 characters), uniqueness check
- **API Endpoints:** POST /auth/register, POST /auth/login, POST /auth/logout
- **Security:** JWT tokens, password hashing with bcrypt

Feature: Product Catalog

- **Screens:** Product Listing, Product Detail
- **Display:** Grid layout, product images, names, prices
- **Search:** By product name
- **API Endpoints:** GET /products, GET /products/{id}, GET /products/search
- **Admin Functions:** POST /products, PUT /products/{id}, DELETE /products/{id}

Feature: Shopping Cart

- **Screens:** Cart View
- **Functions:** Add product, remove product, update quantity

- **Persistence:** Database storage per authenticated user
- **API Endpoints:** GET /cart, POST /cart/items, PUT /cart/items/{id}, DELETE /cart/items/{id}

Feature: Checkout Process

- **Screens:** Checkout Form, Order Confirmation
- **Data Collected:** Shipping address (name, address, city, zip)
- **Process:** Validate input, create order, clear cart, show confirmation
- **API Endpoints:** POST /orders, GET /orders/{id}

Feature: Admin Panel

- **Screens:** Product Management, Order List
- **Functions:** Add/edit/delete products, view all orders and their statuses
- **Access Control:** Admin role required
- **API Endpoints:** Admin-prefixed endpoints with role validation

2.5 Acceptance Criteria

AC-1: Successful User Registration

- Given I am a new user
- When I enter valid email and strong password
- And confirm password matches
- And click "Create Account"
- Then my account should be created
- And I should be automatically logged in
- And redirected to the homepage

AC-2: Product Purchase Flow

- Given I am logged in as a customer
- When I add a product to my cart
- And proceed to checkout
- And enter valid shipping information
- And place the order
- Then I should see order confirmation
- And my cart should be empty
- And the order should appear in admin panel

AC-3: Admin Product Management

- Given I am logged in as an administrator
- When I add a new product with valid details
- And save the product

- Then the product should appear in customer product listing
- And be available for purchase